

THE ROLE OF INFORMATION SYSTEMS IN PUBLIC REPORTING - A SYSTEMATIC LITERATURE REVIEW

Abstract

Background: *With the advancement of digitalization, health information systems are becoming increasingly essential in healthcare, facilitating the collection, analysis, and transformation of data from various sectors to foster quality, relevance, and timeliness for informed decision-making. Previous studies indicate that public reporting has, overall, contributed to improvements in healthcare quality and facilitated user choice, including enhanced patient satisfaction and clinical outcomes in pathway theory. However, few studies have provided a comprehensive analysis of the effectiveness and limitations of public reporting system.*

Methods: *A literature search was conducted on five electronic databases: PubMed/Medline, PsycInfo, EMBASE, CINAHL, and Web of Science.*

Results: *This review includes findings from a total of 60 research articles. Internet-based public reporting become predominant after year 2015. The purpose for public reporting is categorized into individual -, organization -, and public reporting -related purposes. Policy, organizational perspectives, and behavioral dynamics have influenced the evolution of public reporting. Meanwhile, during the reporting practice, existing systems fail to adequately meet the needs of individual users. Discrepancies in data consistency and accuracy persist across different systems, highlighting the importance of an interoperable data model that facilitates seamless data exchange across heterogeneous systems.*

Conclusions: *This systematic literature review examines public reporting from a global perspective, analyzing its evolution, impact, and future potential. By exploring the relationships between public reporting systems, data components, and key stakeholders, this study provides valuable insights into the effectiveness of public reporting in driving healthcare transparency and quality enhancement. The main contributions to this work are as follows: (1) to provide the up-to-date review of public reporting in health care from a global perspective; (2) identifying key trends, challenges, and gaps in digital public reporting systems, summarizing the current state of knowledge within an integrated framework; and (3) to reduce the complexity of public reporting by offering structure and clarity.*

Keywords: *Health information system, public reporting, digitalization, health quality, systematic literature review.*

1. Introduction

The digitalization of healthcare has long been regarded as a transformative force within the healthcare industry, and the pandemic is thought to have accelerated this process (Golinelli et al. 2020). In this context, health information systems play a crucial role by collecting, analyzing, and ensuring the quality, relevance, and timeliness of data from the health sector and other relevant sources. These systems then transform the data into actionable information that supports health-related decision-making (World Health Organization 2008). Furthermore, systematic literature reviews adopt clearly defined methodological protocols which enable other researchers to replicate the process and validate the findings (Shaheen et al. 2023). One key approach to enhancing transparency and accountability in healthcare is public reporting, which involves making healthcare quality information publicly accessible to consumers (Kast et al. 2023). Building on this, the primary objective of this study is to provide the up-to-date review of public reporting from the literature on the public reporting in healthcare through a Systematic Literature Review (SLR) (Alavi and Joachimsthaler 1992; Alavi and Leidner 2001). By analyzing public reporting across various disciplines, this study not only synthesizes existing knowledge but also offers valuable insights for scholars, policymakers, and practitioners, contributing to the ongoing discourse on healthcare transparency and quality improvement.

Public reporting of comparative information on health care quality of physicians, hospitals, and health plans through “report cards” is hailed as a plausible way to improve health care (Arrow 1978; Stiglitz 1989). Report cards may be defined as standardized, publicly released reports on quality of care (Epstein 1995). Typically, they are used to make comparisons of performance amongst individuals or organizations (Longo et al. 1997), over time or against defined standards of care. In the literature review, we found that current studies have explored a variety of perspectives on public reporting, it shaped by a combination of societal needs and technological advancements (Busse et al. 2019), involving diverse stakeholders (Martin M et al. 2013; Van Wilder et al. 2021), and policies (Han and Park 2021). It aims to enhance transparency, facilitate informed decision-making in provider selection, drive continuous quality improvement, and promote accountability in healthcare delivery (Busse et al. 2019).

Previous research has been done mainly in three distinct pathways: the selection pathway, the change pathway and reputation pathway. In the selection pathway, consumers compare public reporting data and choose high-quality providers over low-quality providers (Berwick, James, and Coye 2003). The selection pathway has been studied in terms of patient choice (Behrendt and Groene 2016), patient satisfaction (Shintani Smith et al. 2021), physician rating website (Lagu et al. 2010). In the change pathway, organizations identify underperforming areas, leading to performance improvement. These pathways are interconnected by providers’ motivation to maintain or increase market share (Berwick et al. 2003). In the change pathway, studies have examined elements

such as transparency mechanisms (Timofeyev, Dremova, and Jakovljevic 2023) and provider-driven changes, including clinical improvements (Behrendt and Groene 2016; Berger et al. 2013; Campanella et al. 2016) and physician's behavior (Frølich A et al. 2007).

Scholars raised the third pathway, public reporting can negatively affect the public image of a provider or an organization, reputation concerns will therefore motivate providers to protect or improve their public image (Hibbard, Stockard, and Tusler 2005), thus earn more market share.

The reviewed articles primarily focus on specific aspects of public reporting. From the individual perspective, studies have addressed patient satisfaction (Derks et al. 2024). From the organizational perspective, research has examined topic such as transparency mechanisms (Duncanson et al. 2024), clinical improvements (Derks et al. 2024), health disparities (Wadhera et al. 2019), incentive policies (Kurtzman ET, Dawson EM, and Johnson JE 2008), quality policies (Van Wilder et al. 2021), racial disparities (Negele et al. 2023), infection indicators (Emmert and Wiener 2017) and incentive structures (Osińska, Favez, and Zúñiga 2022). Studies on public reporting system have focus on system development (Theis et al. 2017), physician rating websites (Davila et al. 2024), website information display (Vaughn et al. 2020), and information presentation formats (Kolb et al. 2018; Lloyd et al. 2021). Other areas of focus include quality indicators (Shintani Smith et al. 2021), data accuracy (Hsu, Hung, and Wang 2019) and data consistency (Berger et al. 2013; Derks et al. 2024). However, few studies have attempted to provide a comprehensive view of the development of public reporting in healthcare.

Research has also explored the “metric-driven harm” caused by performance measurement systems (Rambur et al. 2013), but some un-intended consequences remain unexplored. For instance, at the individual level, inaccurate reporting information can lead to patient anxiety (Duncanson et al. 2024). Some physicians may avoid high-risk surgical patients in an effort to achieve better public evaluations, physicians and hospitals may focus more on achieving specific “target rates” rather than providing personalized care based on clinical judgment and patient needs (Werner and Asch 2005). At the organizational level, the introduction of public reporting imposes significant financial burdens on hospitals (Lagu et al. 2019). Additionally, reporting bias, data distortion (Martin M et al. 2013), and misuse **may occur** due to inaccurate outcome data (Lagu et al. 2019). A deeper understanding of these issues is crucial to **fostering** that public reporting systems truly improve **healthcare** quality rather than introduce new inefficiencies.

This article aims to answer the following research questions:

RQ1 What types of public reporting systems and data components of the public reporting are used and studied?

RQ2 What are the origin countries and key user groups of public reporting systems identified in the reviewed literature?

RQ3 What are the main purposes of the reviewed studies on public reporting systems?

RQ4 What are the key findings of the reviewed studies on public reporting systems?

Previous literature reviews have primarily examined public reporting from a single perspective, limiting their ability to capture its broader implications. In contrast, this study adopts a comprehensive, global perspective, incorporating multiple viewpoints to provide a more holistic analysis. By synthesizing and reporting on the key research focuses of 15 literature articles, it identifies critical gaps in existing reviews. To enhance clarity, we categorize public reporting into three dimensions: individual, organizational, and public reporting.

At the individual level, our analysis finds that differences in health status, cognitive ability, and access to information among groups can affect how public reporting influences healthcare outcomes. These differences suggest that public reporting has the potential to reduce both general health and racial disparities.

At the organizational level, earlier studies have emphasized the importance of considering institutional differences when developing policies. For example, financial incentives play a crucial role in improving healthcare quality (Frølich A et al. 2007). Our work highlights those variations between organizations are a critical factor in determining the success of public reporting initiatives.

At the public reporting level, the presentation of data is key to consumer decision-making. To avoid misinterpretation—where performance data might be seen as a direct measure of quality—it is essential to verify data through mandatory reviews and validation processes (Kurtzman and Greene 2016), (Lucet et al. 2013). In addition, quality indicators must accurately reflect service standards to allow for both international comparisons and internal improvements (Osińska et al. 2022).

Beyond these dimensions, our study also addresses unintended consequences, data integrity, and system factors. We suggest that presenting quality data more comprehensively can improve public understanding. Maintaining data authenticity and consistency is essential for credibility. Furthermore, training staff in accurate data entry and standardizing system use can help overcome potential challenges.

By applying a categorization framework to align the purpose and consequence of public reporting, our study not only contributes to the academic literature but also provides practical insights for researchers, policymakers, and healthcare practitioners in guiding future policy development and system optimization.

The remainder of this paper is organized as follows: Section 2 outlines the methodology for planning and conducting the literature review. Section 3 presents the findings corresponding to each research question, summarizing the current state of knowledge within an integrated framework reduce the complexity of public reporting by offering structure and clarity. **Section 4 outlines potential directions for future research. Section 5 presents the contributions and research implications. Section 6 provides a concluding summary and highlights the limitation.**

2. Methods

2.1 Searching Process of the Systematic Review

A research literature review is defined as "a systematic, explicit, and reproducible approach for identifying, evaluating, and synthesizing the existing body of completed and documented work produced by researchers, scholars, and practitioners" (Fink 2013), when applied rigorously, this method fosters comprehensive synthesis, aiming at providing a clear and reliable foundation for future research and practice. The primary aims of such reviews are to identify emerging research areas, facilitate theory development, address gaps in saturated fields, and categorize studies within specific research streams (Okoli 2015; Vom Brocke et al. 2015; Webster and Watson 2002). This review follows the guidelines proposed by **Webster and Watson (2002)**. The assessment is complemented by an external evaluation of the outcomes at each juncture.

Articles were retrieved from five academic databases: PubMed/Medline (National Library of Medicine), PsycInfo (American Psychological Association), EMBASE (Excerpta Medica Database, Elsevier), CINAHL (Cumulative Index to Nursing and Allied Health Literature, EBSCO), and Web of Science (Clarivate Analytics).

Reviewers are constrained to evaluating studies published in languages they can proficiently understand or those accessible via scholarly databases (Fink 2019). A language restriction was set to English to **facilitate** easy querying and referencing of the results by a global audience.

Access to these databases was obtained via the University of Turku facilities, ensuring that the user was logged in to minimize the amount of ignored articles due to access issues. The number of search queries for each database was recorded to verify the accuracy of each subsequent search step.

EndNote software was employed to manage the article search and facilitate the analysis process. To promote comprehensive article retrieval and reduce the risk of missing items during the search process, the number of search results, downloads from each database, and imports into Endnote software were all monitored. In the analysis phase, duplicate entries were first removed using EndNote software, followed by manual checks for duplicates and title screenings to enhance accuracy. The analysis phase started with the elimination of 1607 duplicates.

University of Turku librarian was consulted about the search terms. The titles and abstracts of articles were searched using the following terms to identify subject-specific literature for the review: (“healthcare,” “health care,”) and (“public reporting,”). The search strategy for each database is outlined in Table 1, and the detailed explanation of these search strategies is provided. The search string was designed for broad coverage to avoid missing any relevant research. No limitations were placed on publication date or study design; however, studies published in languages other than English were excluded.

Initially we found 2690 publications in total, of which 746 in Pubmed, 563 in WoS, 411 in CINAHL, 830 in Embase and 140 in PsycInfo. Fink recommends electronic tools such as reference managing software to create a “virtual filing cabinet” (Fink 2019).

Database	Search terms
PubMed (746)	("healthcare"[Title/Abstract] OR "health care"[Title/Abstract]) AND "public report*" [Title/Abstract]
PsycInfo (140)	(TI healthcare OR AB healthcare OR TI "health care" OR AB "health care") AND (TI "public report*" OR AB "public report*")
EMBASE (830)	(healthcare OR "health care"):ab,ti AND (("public report*"):ab,ti)
CINAHL (411)	(TI healthcare OR AB healthcare OR TI "health care" OR AB "health care") AND (TI "public report*" OR AB "public report*")
Web of science (563)	(TI=(healthcare OR "health care") OR AB=(healthcare OR "health care")) AND (TI="public report*" OR AB="public report*")

Table 1. Databases and search terms.

2.2 Screening Process of the Systematic Review

In the screening process, first abstracts, keywords, and titles from the five databases of the found articles were screened based on the previously mentioned search terms. Further screening of the 1083 articles was conducted using inclusion and exclusion criteria in a two-step process. In the fourth step, practical screening criteria were applied. The literature review initially screened the titles and abstracts of the 1,083 articles to determine their relevance to public reporting. Selected articles were then subjected to forward and backward citation tracking to **enhance** the comprehensiveness of the literature search.

After reviewing the titles and abstracts, 993 records were excluded for not matching the inclusion criteria (i.e., studies with the main topic (title/abstract/keywords) containing public report, studies available in English). Our search was confined to English-language publications from 2005 to 2024, as recommended by the Cochrane Rapid Reviews guidance (Garritty et al. 2021). This timeframe was selected based on preliminary searches indicating that the majority of pertinent articles were published during this period.

During the screening process, the titles of the 1,083 retrieved articles were reviewed to determine whether they contained the keyword “public reporting.” If the title is unclear, the abstract was read to assess whether the study focused on public reporting. Articles that merely mentioned public reporting—such as those recommending attention to public reporting or using public reporting data to examine unrelated research—were also excluded.

Citation chaining was conducted to further reinforce the comprehensiveness of the article search. The inclusion criteria and exclusion criteria for each database is summarized in Table 2. By further not fulfilling the inclusion criteria at the analysis level of full text, 8 articles were excluded, leaving 82 studies.

Inclusion criteria (IC)
IC1: Studies published in English
IC2: Studies with the main topic (title/abstract) includes public report
IC3: Publications from 2005 to 2024
IC4: Full-texts availability of studies
IC5: Studies published in journals and conferences papers
Exclusion criteria (EC)
EC1: Studies whose main topic is not public reporting (i.e., even if public reporting is mentioned in the abstract, the article is not primarily about public reporting)
EC2: Full text unavailable
EC3: Studies that are not peer-reviewed articles (books, letters, editorial, dissertation) and literature review articles
EC4: Publications outside the period of 2005–2024 were excluded

Table 2. *Inclusion and exclusion criteria.*

The types of articles were further screened. The remaining 82 articles were reviewed in full to determine whether they met inclusion criterion 5 (i.e., studies published in journals and conference proceedings). Fifteen literature reviews and 8 non-peer-reviewed (e.g., books, editorial, dissertation, letters) were excluded. Conceptual papers published in peer-reviewed journals were included as they contribute valuable theoretical insights. Notably, the excluded articles were organized into a separate subfolder, categorized according to the set inclusion criteria they did not fulfill (Okoli 2015).

Three articles were excluded from the analysis of public reporting material because their research focused on general discussions of public reporting without providing any specific data relevant to our analysis. At the end, 60 academic articles were selected as the final sample. Figure 1 illustrates the phases of article retrieval and the number of articles included.

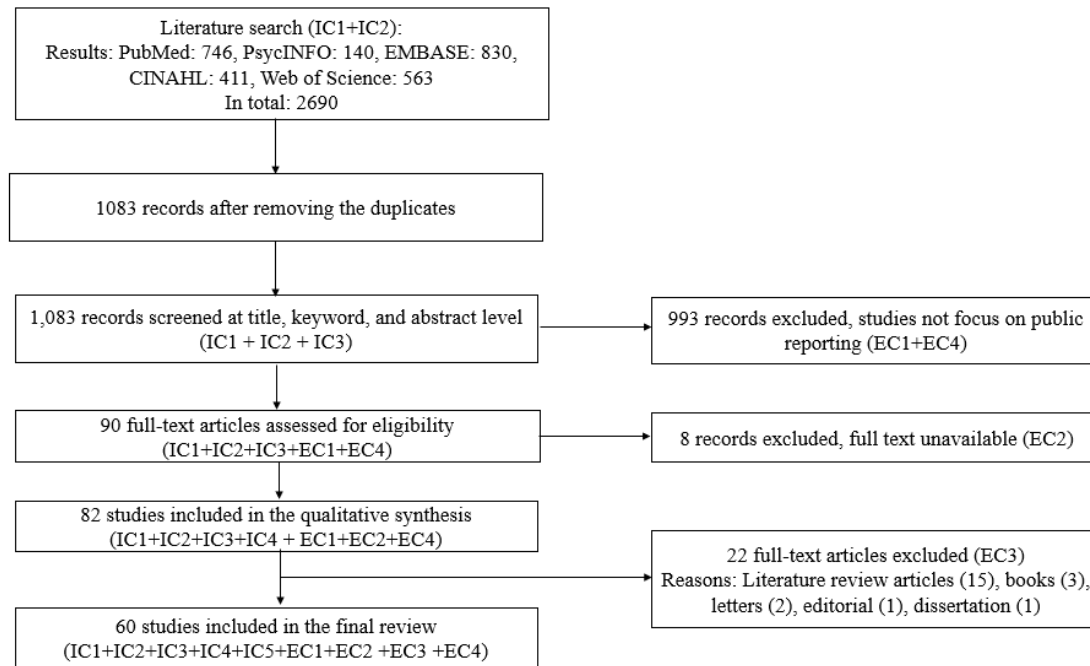


Figure 1. Literature search steps. IC: inclusion criterion; EC: exclusion criterion.

2.3 Data Extraction, Coding and Analysis of the Systematic Review

Coding was not limited to article titles and abstracts but was derived from the full content of the reviewed studies. To analyze the selected articles, definitions, research methods, theoretical frameworks, and study contexts were extracted from the abstract, introduction, methods, and results sections of the reviewed articles. Information was summarized and coded into Excel spreadsheets for further analysis.

The content analysis on RQ1 and RQ2 has been done through searching for 'structures and patterned regularities in the text' (Myers 2019). Looking for the keywords or phrases indicate the answer to the research question from the abstract, introduction, method, discussion implication and conclusion, then extracting relevant text segments as supplement information for RQ1 and RQ2.

For RQ1, the level of analysis was determined using keyword queries such as "system" and "data." For RQ2, coding was based on a predefined set of concepts, including "purchaser," "provider," and "policy maker" (Dehmer et al. 2014). Country information was extracted by reviewing the full text. In cases where the study's geographic focus was not explicitly stated, data sources, research institutions, and author affiliation was considered as a supporting indicator, but final classification was based on a full-text review to accurately identify the country of focus.

However, the diversity of user roles posed challenges in classification, certain studies featured multiple roles within the same research context. For example, interviews included both individuals and employees, with some employees representing both their organizations and themselves. This dual identity introduced potential classification biases, resulting in articles being assigned to multiple categories.

For addressing RQ3 and RQ4, Open, Axial, and Selective (OAE) coding (Alhassan et al. 2023; Corbin and Strauss 1990; Seidel and Urquhart 2013; Strauss 1987) coding due to these established application in information system research.

According to Strauss and Corbin (Corbin and Strauss 1990) concepts that appear similar are grouped under a higher-order, more abstract category. The open coding includes the appeared category of helping individual making health decision, organization make improvement and improve transparency. And the category of purchaser, provider and policy maker. But these category fails to draw the relationship between the individual, organization and the system. After the data was broken down into concepts and categories with properties and dimensions, the next step was to reconstruct the data into a relational form (Corbin

and Strauss 1990). We organized the categories of individual, organization, and information system according to the purposes and results from the reviewed articles. Each category contains sub-categories, which serve as dimensions to provide structure to the findings.

We reviewed the terms within each sub-category, ensuring that results from previous research were explained, and similar results were grouped together. The coding framework helped minimize the risk of overlooking articles, following open and axial coding, we transitioned from an author-centric to a concept-centric approach, as presented through the concept matrix (Webster and Watson 2002).

Through selective coding, the core category, which represents the central phenomenon, was re-organized. This process allowed the analyst to develop a coherent narrative about the study area (Corbin and Strauss 1990). By using this method, the reviewed article can be present as detail as possible. The identified key themes based on the RQs were reported in Table 3. The identified sub-categories in RQ3 and RQ4 were visualized in Figure 3, which informed the results presented in Section 5.

Research Questions	Key Themes
(1) What types of public reporting systems and data components of the public reporting are used and studied	Public reporting systems and data component of the systems based on RQ1
(2) What are the origin countries and key user groups of public reporting systems identified in the reviewed literature	- Key user groups of public reporting systems based on RQ2 - Countries of origins based on RQ2
(3) What are the main purposes of the reviewed studies on public reporting systems	- Individual-related purposes on public reporting system based on RQ3 - Organization-related purposes on public reporting system based on RQ3 - Public reporting -related purposes on public reporting system based on RQ3
(4) What are the key findings of the reviewed studies on public reporting systems	- Individual-related consequences on public reporting system based on RQ4 - Organization-related consequences on public reporting system based on RQ4 - Public reporting -related consequences on public reporting system based on RQ4

Table 3. Key themes coded based on the Research Questions.

3. Results

3.1 Bibliographical analysis

In the bibliographical analysis we focused on two key indicators: the country that was object of the reporting, and the number of publications per year worldwide.

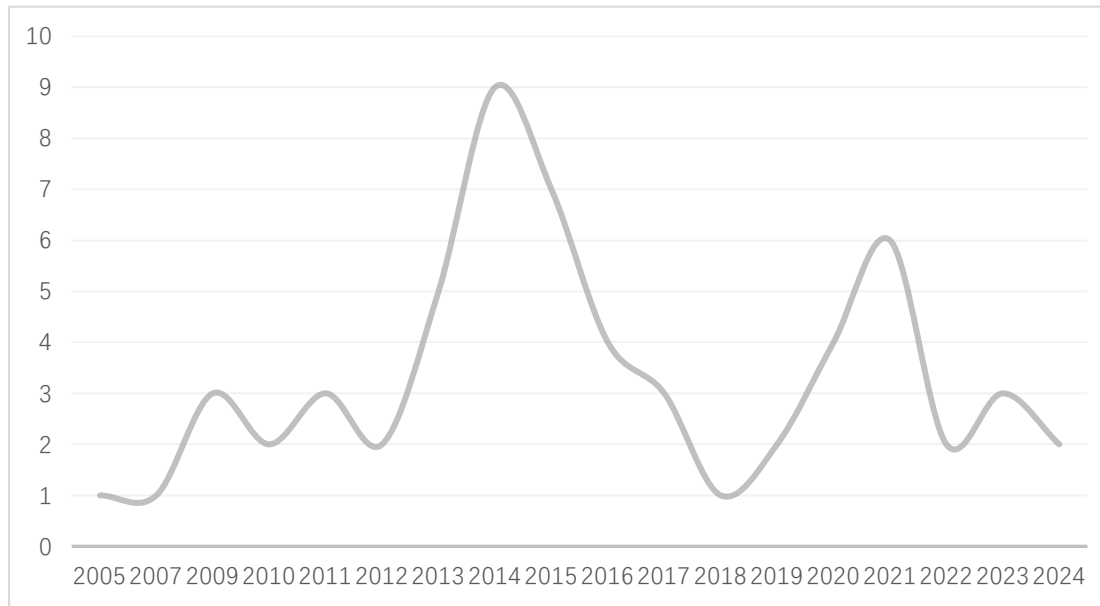


Figure 2. *Publications per publication year.*

The data reveal two significant peaks in public reporting research, occurring in 2014 and 2021. The first peak in 2014, with nine published studies, marks a high point in public reporting research. However, from 2015 onward, the number of studies steadily declined. As public reporting transitioned to internet-based platforms, the academic focus on traditional reporting formats diminished, contributing to this decline.

The second peak in 2021 reflects the profound impact of global events, particularly the COVID-19 pandemic, on the academic community. The pandemic renewed interest in healthcare systems and public reporting, leading to a resurgence in research activity.

Between 2023 and 2024, fluctuations in research output may be attributed to shifts in research priorities and resource allocation. Additionally, part of this variation may result from incomplete data, as studies from 2024 may still be in progress or awaiting publication. Overall, this trend highlights the evolution of public reporting research and its dynamic relationship with technological advancements and major global events, offering valuable insights for future studies.

As was shown in Table 4, after year 2015 the number of countries whose public reporting was studied are the United States (n=26), China (n=3), Germany (n=3), Europe (n=2). Before year 2015, the country who studied public reporting becomes to the United States (n =16), Germany (n=5), Australia (n=3), China (n=1), Canada (n=1), Belgium (n=1), Korea (n=1).

Country studied	Frequency	Key national policies that might have an effect on publication amounts
United States	43	Affordable Care Act (ACA) (Han and Lee 2021), pay-for-performance programs (Crews et al. 2016), mandatory vaccination policies (Paris et al. 2013), mandatory public reporting system for Healthcare-associated infections (HAIs) (Edmond and Bearman 2007).
Germany	5	Since 2005, German hospitals have been required to regularly publish structured quality reports (Emmert and Wiener 2017).
Australia	3	The use of pay-for-performance (P4P) (Duncanson et al. 2024).
Europe	2	Healthcare-associated infections (HAIs) (Martin M et al. 2013), "patient-centred care" strategies, and pay-for-performance (P4P) approaches (Schlesinger, Grob, and Shaller 2015).

Table 4. Countries where the studied public reporting systems were operating, and relevant policy actions.

The bibliographic analysis revealed that the United States ($n = 43$) and Germany ($n = 5$) had the highest number of studies. Several articles focused on transparency (in the U.S.) or structured implementation (in Germany), reflecting how policy environments may influence research emphasis. To maintain clarity, Germany is listed separately from studies referencing Europe in general. This categorization follows the source articles and contributes to the consistency of reporting.

3.2 System types and their input and output

3.2.1 Types of Public Reporting Systems studied

The term "public reporting" encompasses the dissemination of information and reports regarding the quality of care provided by health and long-term care providers, intended for use by patients and other stakeholders. This information is typically accessible through online platforms, brochures, and other printed materials (Kumpunen et al. 2014). Our analysis too indicates that public reporting systems are called with very various names. This also makes their comparison and analysis difficult, as well as finding all relevant articles.

A key distribution goes between digital (computerized) and analog (paper-based) public reporting outputs. By reading the abstracts, introductions, methods and results sections for descriptions, then checking the keywords “internet”, “online”, “web”, “printed”, “paper” and “brochure” in the reviewed article, the majority of public reporting discussed in the reviewed studies are digital (computerized), with the number of 54 (90%). Early studies, particularly those conducted before 2015, focused on analog (paper-based) public reporting, with the number of 3 (5%), which highlights that the Internet has become the predominant medium for studying public reporting. Older studies often centered on analog (paper-based) methods but lacked clarity in their descriptions, whereas newer studies tended to emphasize digital approaches without explicitly detailing their methodologies.

The key names found in our analysis, that we interpret as being at core public reporting systems, even though they might have other functions too, are presented in Table 5.

Type of System	Frequency	Examples of systems	Articles
Public Reporting Systems	7	• Report cards (e.g., hospitals, nursing homes) • Facebook & Yelp • Ascension Health	(Geraedts et al. 2018), (Emmert and Wiener 2017), (Emmert et al. 2017), (Sander et al. 2015), (Konetzka et al. 2015), (Lagu and Greaves 2015), (Fakih et al. 2016)
Monitoring and prevention Systems	6	• Healthcare-associated infections (HAIs) • Public Health Automated Case Event Reporting (PACER) • Public health systems	(Martin M et al. 2013), (Hebden 2012), (Mazor KM, Dodd KS, and Kunches L 2009), (Edmond and Bearman 2007), (Mishra et al. 2020), (Chae et al. 2022)
Comparative Reporting Systems	5	• Hospital comparison • Home health comparison • Nursing home comparison • Medical hospital comparison	(Lagu et al. 2019), (Werner, Konetzka, and Polsky 2016), (Werner et al. 2009), (Emmert et al. 2013), (Emmert and Meier 2013)
Patient-Centered Systems	4	• Electronic Health Records (EHR) • Electronic Medical Record (EMR)	(Prang et al. 2021), (Poldrugovac et al. 2022), (Tang, Liu, and Zhang 2017), (Walters et al. 2014)
Claims-Based Systems	4	• All-Payer Claims Databases (APCD) • Wisconsin Collaborative for Healthcare Quality (WCHQ) and Wisconsin Health Information Organization (WHIO)	[41], (Han and Lee 2021), (Crews et al. 2016), (Toussaint, Shortell, and Mannon 2014)
Performance Measurement/Registries	4	• Performance measurement systems • Australia and New Zealand Dialysis and Transplant Registry (ANZDATA) • Canadian Institute for Health Information (CIHI) • National Cardiovascular Data Registry (NCDR)	(Duncanson et al. 2024), (Rambur et al. 2013), (Poldrugovac et al. 2022), (Dehmer et al. 2014)

Hospital and Institutional Systems	4	• Hospital Information System (HIS) • National Survey of Substance Abuse Treatment Services (N-SSATS)	(Du, Wang, et al. 2015), (Yang et al. 2014), (Vaughn et al. 2020), (Saghafian and Hopp 2020)
Rating Systems	4	• Five-star rating systems • Physician rating websites	(Davila et al. 2024), (Werner et al. 2016), (Emmert et al. 2013), (Emmert and Meier 2013)

Table 5. *Types of Public Reporting Systems in Reviewed Articles.*

The most frequently studied system is the public reporting system (n=7), which includes hospital report cards, provider report cards, and nursing home report cards. Ranked alongside others in second place are the monitoring and prevention system (n=6). Ranked third is the comparative system, which has gained prominence due to the growth of the internet and the expanded use of user-generated data. Research suggests that physician rating websites offer patients a new way to provide feedback and obtain information about physician performance, and their influence will continue to grow as public awareness grows (Emmert et al. 2013). Among these ratings, female physicians are rated significantly higher than male physicians, and older patients and those with private health insurance generally give higher ratings. Most patient reviews tend to be positive, with about two-thirds rating them “very good” (Emmert and Meier 2013).

With the launch of Nursing Home Compare public reporting, most publicly reported nursing quality indicators have improved (Werner et al. 2009). Public reporting mechanisms, such as Medicare’s Hospital Compare and Physician Compare websites, are driving greater transparency (Lagu et al. 2019). Overall, star rating systems have significantly affected nursing home market share, with consumers preferring higher star-rated nursing homes (Werner et al. 2016).

This trend includes the application of health-related data in various contexts, reflecting the increasing reliance on digital platforms and data-driven decision-making. The plentitude of system types studied under main title “public reporting system” underlines the need for further conceptualization of public reporting systems, a task we too perform for our part in this article.

3.2.2 Key data sources and output forms for public reporting

Previous studies have identified issues such as reporting bias, data distortion (Martin M et al. 2013), and misuse due to inaccurate outcome data (Lagu et al. 2019). As any information system, public reporting systems require input data to generate output. This paper classifies studies focusing on performance and quality indicators as input studies since they are conducted before data are released publicly, while studies that analyze publicly available data are categorized as output.

From an input perspective, patient information (n=7) is the primary source, followed by quality performance data. Research in this area mainly concentrates on quality and performance indicators. Public reporting tends to enhance healthcare quality by drawing the attention of hospital leadership to the data, which in turn encourages hospitals to address the deficiencies highlighted in these reports (Hafner JM et al. 2011).

In terms of indicator selection, disclosing structural and process measures has proven more effective in improving hospital quality than reporting infection rates directly (Martin M et al. 2013). For instance, public reporting has notably improved process measures for acute myocardial infarction (AMI) care, such as timely reperfusion rates, especially in hospitals with limited resources. However, while these process indicators have seen improvements, outcome indicators like mortality rates have not changed significantly, suggesting that further refinement of quality indicators is necessary (Chae et al. 2022).

Data/Information	Frequency	Output data
Patient-related Data	7	Electronic Health Records (Smith et al. 2023), nursing home report cards (Konetzka et al. 2015), HRC (Hospital Report Cards) (Werner and Asch 2005), (Emmert and Wiener 2017), patient demographics (Tang et al. 2017), infection prevention scores (Duncanson et al. 2024), Patient-Reported information (PRI) (Schlesinger et al. 2015).
Quality & Performance Data	6	Risk-Adjusted Outcome Indicators (Bayindir and Schreyögg 2023), quality indicators (Bayindir and Schreyögg 2023), (Rambur et al. 2013), patient quality indicators (Ellis et al. 2020), prescription quality indicators (Du, Zhang, et al. 2015), five-star rating data (Werner et al. 2016).
Healthcare Institution Data	1	Data from Centers for Medicare and Medicaid Services (CMS) (Saghafian and Hopp 2020).

Table 6. *Data (input) components of public reporting in the reviewed articles.*

As shown in Table 7, most output data in these studies consist of quality information (n=7) and performance information (n=6). With patient-centered system are frequently cited, public reporting developers need to focus more on user experience, ensuring that reporting formats better accommodate diverse user groups (Shaller, Kanouse, and Schlesinger 2014). Furthermore, research indicates that while patients generally support the public reporting of healthcare performance data, some experience anxiety—especially when they have no option to change treatment centers. Patients prefer that the data be presented positively and accompanied by clear explanations and educational content, which helps them understand the information and make informed decisions (Duncanson et al. 2024).

Information	Frequency	Examples of data/information
Quality data	7	Acute myocardial infarction (AMI) (Chae et al. 2022), percutaneous Coronary Intervention (PCI) (Wadhera et al. 2019), kidney disease quality data (Duncanson et al. 2024), publicly available performance data (Geraedts et al. 2018; Toussaint et al. 2014), hospital beds [72], full-time equivalent staff (Han and Lee 2021), physician information (Geraedts et al. 2018).
Hospital performance data	6	Antibiotic prescription percentage ranking (Du, Zhang, et al. 2015), chronic condition management (e.g., diabetes care) (Snowden AM et al. 2012; Yonek, Mittler, and Hasnain-Wynia 2014), (Shaller et al. 2014), patient satisfaction scores (Emmert et al. 2017), mortality rates (Crews et al. 2016).
Healthcare Institutions and Care Quality Data	6	CLCs (Community Living Centers) care quality data (Pimentel et al. 2021), PICU (Pediatric Intensive Care Unit) performance data (Natale et al. 2011), Healthcare worker (HCW) flu vaccination coverage (Johnson and Talbot 2011), prescription behavior data (Du, Wang, et al. 2015), prescription quality indicator data (Du, Zhang, et al. 2015), prescriber details (name, work unit, time of prescriptions) (Johnson and Talbot 2011; Martin M et al. 2013).
Healthcare-associated infections (HAI)	4	Healthcare-associated infections (HAIs) data (Edmond and Bearman 2007; Fakih et al. 2016; Hebden 2012; Mazor KM et al. 2009).
Operations and Resource Management Data	3	Elective surgery wait times (Prang et al. 2021), operating expenses (Han and Lee 2021), 30-day readmission rates (Tang et al. 2017), flu vaccination rates (staff behavior data) (Paris et al. 2013).07/10/2026 12:53:00
Public Data	2	Publicly available claims data (Toussaint et al. 2014), HHC (Home Health Compare) public reporting data (Jung K, Shea D, and Warner C 2010), public database systems (e.g., databases maintained by SAMHSA) (Vaughn et al. 2020).

Website information	2	Medicare Nursing Home Compare (Davila et al. 2024), Care website (accessstocare.va.gov) (Werner et al. 2009).
Patient experience data	2	User-generated data (Emmert et al. 2013), patient experience data (Lagu et al. 2019).

Table 7. *Information (output) components of public reporting in the reviewed articles.*

3.3 Stakeholders of Public Reporting Systems Identified in the Reviewed Literature

There is an ongoing debate about the effectiveness of existing Health Technology Assessment (HTA) (Fehr et al. 2017). Beyond methodological concerns, discussions emphasize the inclusion and education of key stakeholders and end-users, the adoption of a systematic approach to prioritizing HTA topics, the integration of real-world data, and the establishment of criteria for disinvestment (Anon 2016). A systematic analysis of user roles within public reporting systems can help identify diverse needs and examine how these variations influence system development.

In the context of public reporting, user roles are traditionally categorized into purchasers, providers, and regulators (Dehmer et al. 2014). However, the emergence of new user roles, seldom considered in previous studies, led to an inductive approach in this section. Notably, the reviewed articles often discuss multiple user roles, with some employing mixed-methods approaches, thereby incorporating multiple users in a single study.

To provide a comprehensive overview of user engagement in public reporting, we have compiled statistical data on user roles based on their representation in the reviewed studies. As presented in Table 8, findings indicate that staff (n=50) are the most frequently studied stakeholder group in public reporting research.

Category	Frequency	Examples of stakeholders
Staff	50	Hospital Administrators (Bayindir and Schreyögg 2023; Edmond and Bearman 2007; Ellis et al. 2020; Han and Lee 2021; Mazor KM et al. 2009; Natale et al. 2011; Poldrugovac et al. 2022; Werner et al. 2009), Infection Prevention Experts (Martin M et al. 2013), (Hebden 2012), Healthcare IT Specialists (Mishra et al. 2020), Nursing Staff (Konetzka et al. 2015), (Rambur et al. 2013), Staff responsible for evaluating the performance of hospitals (Paddock 2014).
Policymakers	33	Public Health Researchers (Smith et al. 2023), (Chae et al. 2022), (Poldrugovac et al. 2022), (Van Wilder et al. 2021), (Han and Lee 2021), (Vaughn et al. 2020), (Saghafian and Hopp 2020), (Mishra et al. 2020), (Ellis et al. 2020), (Lagu et al. 2019), (Johnson and Talbot 2011), (Werner et al. 2009), (Mazor KM et al. 2009), (Edmond and Bearman 2007), Health Bureau Administrators (Du, Wang, et al. 2015), (Prang et al. 2021).
Patients	17	Hospital Patients (Jung et al. 2016), Hospital Report Cards (HRC) Users (Emmert and Wiener 2017), (Sander et al. 2015), Family Members (Sander et al. 2015), (Lihn et al. 2015), (Lagu and Greaves 2015), (Konetzka et al. 2015), (Du, Wang, et al. 2015), (Shaller et al. 2014), (Christianson et al. 2010), Elderly Patients with Chronic Conditions (Christianson et al. 2014), Community Patients (Jung et al. 2016).
Organizations	11	Health Insurance Companies (Negele et al. 2023), (Yonek et al. 2014), (Yang et al. 2014), (Toussaint et al. 2014), Professional Organizations for Developing and Evaluating Quality Reports (Dehmer et al. 2014), (Rambur et al. 2013), (Rhoads, Konety, and Dudley 2009), Health Department (Yang et al. 2014; Yonek et al. 2014), Infection Control Team (Fakih et al. 2016), Canadian Nursing Center Government Regulatory Agency (Poldrugovac et al. 2022).
Family member	8	Patient and family member (Christianson et al. 2010; Du, Wang, et al. 2015; Konetzka et al. 2015; Lagu and Greaves 2015; Lihn et al. 2015; Mazor KM et al. 2009; Sander et al. 2015; Shaller et al. 2014).
Analysts	4	Data Analysts (Crews et al. 2016), (Snowden AM et al. 2012), (Hebden 2012), Public Reporting Developers (Shaller et al. 2014).

Table 8. *Classification of Stakeholders in Reviewed Articles.*

The primary aim of public reporting being to aid users in making healthcare decisions, there is limited research focusing on patients (n=17).

It is also noteworthy that, at the individual level, patient family members were included in studies, with findings suggesting that their involvement enhances clinical outcomes. However, system use by this role remains underdeveloped.

In the review articles, staff members play a significant role. The employees in these review articles come from various institutions, including hospital nurses, hospital administrators, data analysts, and data collectors from regulatory agencies. Given these diverse roles and institutional backgrounds, the objectivity and subjectivity of their perspectives require further examination and analysis to understand how their positions influence public reporting and decision-making.

3.4 Main Purposes of the Reviewed Studies

3.4.1 Framework for Understanding Purposes

Selection pathway includes the individual and public reporting system, change pathway includes the organization and public reporting system, reputation pathway include organization and individual. An categorization framework to present a holistic view of these factors is developed, which could guide future research in the field (Huang et al. 2023). Purpose resulted in three main themes: **individual-related purposes, organization-related purposes, and public reporting -related purposes**. For individual-related purposes, we included aspects such as personal behaviors and health disparities. For organization-related purposes, we focused on organizational behaviors and attitudes. For public reporting -related purposes, we examined quality indicators, performance indicators, data

utilization, and system usage. A concept matrix was used to synthesize findings across studies, following a concept-centric approach, which was presented in Appendix A (Salipante, Notz, and Bigelow 1982; Webster and Watson 2002).

3.4.2 Individuals-related Purposes

Individual Choice

When relates to individual's choice, studies have explored the impact of public reporting on healthcare quality (Hussey, Luft, and McNamara 2014), particularly examining the potential positive and negative effects of report cards on the behaviors of doctors, hospitals, and patients (Werner and Asch 2005). Other studies aim to understand how public reporting influences individuals' healthcare choices; for example, social media review systems (Lagu and Greaves 2015) have significantly enhanced individuals' understanding of healthcare performance. Research has also investigated attitudes and preferences related to public reporting data among patients in community living centers (Davila et al. 2024). Additionally, some studies have focused on differences in public reporting preferences between HRC users and non-users (Emmert and Wiener 2017). Systems such as the nursing home 5-star rating system (Werner et al. 2016), quality ratings for home care institutions (Jung et al. 2016), and physician rating websites (Emmert et al. 2013) have all been examined for their impact on personal healthcare decision-making.

Health Literacy **Information Processing**

Additionally, studies have analyzed how different cognitive backgrounds (Emmert et al. 2013) and decision-making contexts (Shaller et al. 2014) affect consumer understanding and decision-making related to public reporting. Factors such as data consistency, report format (text or chart), and the use of confidence intervals have been examined for their influence on consumer comprehension and decision-making (Mazor KM et al. 2009). Research also examined the impact of various presentation formats, such as symbols and charts, to identify methods that improve user comprehension (Sander et al. 2015). Studies have considered the impact of quality reports on the care of patients with advanced dementia (Konetzka et al. 2015), and factors affecting elderly chronic patients' perceptions of physician quality (Christianson et al. 2014). Community studies have explored ways to use public reporting to reduce health disparities and improve awareness among minority groups (Yonek et al. 2014).

3.4.3 Organization-related Purposes

Organization Attitude

Studies on organizational attitudes have often included physicians, as their behaviors and attitudes can represent those of the healthcare institution they work for. Most research has focused on policies and data within public reporting systems. Studies have examined the perspectives of healthcare system stakeholders on the quality policies related to public reporting (Van Wilder et al. 2021), as well as the attitudes of pediatric intensive care (PICU) directors towards benchmarking, public reporting, and pay-for-performance policies (Natale et al. 2011). Physician attitudes toward patient experience data made public online (Lagu et al. 2019), the availability of information, along with perceived value and attitudes of general practitioners to the behavioral intentions (Du, Zhang, et al. 2015), the perspectives of European infection control stakeholders on reporting HAI data (Martin M et al. 2013), and the attitudes of community living centers' (CLC) caregivers toward public reporting systems (Lloyd et al. 2021; Pimentel et al. 2021) have also been analyzed.

The phenomenon of “metric-driven harm” has been analyzed through case studies to examine its effects on healthcare outcomes (Rambur et al. 2013), while the financial costs and administrative burdens of PCI result reporting programs on hospitals and physicians have been investigated (Wadhera et al. 2019).

Organizational Policy **Key National Policies**

This section explores how transparency within collaborative projects contributes to quality improvement and reduced mortality rates. Studies have offered practical recommendations for enhancing quality competition in Germany (Negele et al. 2023) and examined the impacts of public reporting transparency on patient choice, hospital investment, social welfare, and healthcare market structures (Lihn et al. 2015; Saghafian and Hopp 2020). The studies further investigate the foundational assumptions of mandatory public reporting and assess its potential effectiveness in improving hospital quality (Edmond and Bearman 2007). Other research analyzes transparency mechanisms, such as how the publication of prescription data affects prescribing behaviors (Du, Wang, et al. 2015). Two public reporting programs were compared for their differing impacts on clinical and perceived quality (Han and Park 2021). Research also examined how public reporting and market competition influence hospitals’ technical efficiency (Han and Lee 2021) and explored the relationship between patient experience and financial incentives to understand how public policy and healthcare practices can better promote patient-centered outcomes and choice (Schlesinger et al. 2015). Specifically, studies have explored how public reporting mechanisms affect general practitioners’ prescribing behaviors, focusing on factors such as information availability, GP attitudes, perceived value, and perceived risk (Du, Zhang, et al. 2015).

3.4.4 Public Reporting System -related Purposes

Quality Indicators

Scholars have examined how public reporting influences individual choices in healthcare, finding that quality indicators play a significant role. Patients often consider structured data, such as survival and infection rates, when making choices (Duncanson et al. 2024).

Studies also explore the impact of public reporting on quality improvement, particularly regarding process indicators like reperfusion rates and mortality rates for AMI patients (Chae et al. 2022), as well as the importance of quality indicators in selecting hospitals and physicians (Ellis et al. 2020).

Other studies focus on how public reporting systems have improved healthcare quality and reduced costs in Wisconsin and California (Toussaint et al. 2014), or whether mandatory public reporting assumptions effectively enhance hospital quality (Edmond and Bearman 2007). Research shows that public reporting can support consumer choice and encourage providers to improve care quality (Christianson et al. 2010) and that it has had positive effects on post-acute care quality in U.S. nursing homes (Werner et al. 2009), cancer surgery wait times (Prang et al. 2021), appropriate antibiotic use in URTI patients (Yang et al. 2014), and flu vaccination rates among healthcare workers (Paris et al. 2013).

Public reporting has too contributed to quality improvement in cardiovascular specialty care (Dehmer et al. 2014). Emerging strategies to improve vaccination rates as a quality and safety indicator for healthcare workers are also being explored (Johnson and Talbot 2011), alongside home healthcare quality improvement efforts (Jung K et al. 2010).

Performance Indicators

Studies have assessed whether performance indicators in long-term care facilities show improvement following public reporting and compared these trends with non-reported indicators (Poldrugovac et al. 2022). Additionally, research has shown that public reporting of healthcare performance data can reveal disparities across insurance plans, thereby encouraging improvement (Rhoads et al. 2009; Snowden AM et al. 2012).

Reputational and financial incentives are central drivers in quality improvement. Comparative studies on provider performance evaluation (Paddock 2014) and the combined effects of public reporting and pay-for-performance models further emphasize how such incentives motivate providers to raise care standards. Similarly, the effectiveness of public reporting in improving operational performance in rural Australian hospitals has been evaluated (Lloyd et al. 2021).

Research has also examined the information valued most by referring physicians in hospital quality reports, such as hospital performance and patient experience data, to better meet physicians' needs (Geraedts et al. 2018). Other studies investigate how public reporting influences physicians at various performance levels (Tang et al. 2017) and whether public reporting can improve German hospitals' quality without financial incentives (Bayindir and Schreyögg 2023).

Finally, research into staff perceptions highlights how publicly reported performance data can shape internal decision-making and organizational responses (Hafner JM et al. 2011).

Data Usage

Research has explored the feasibility of using state-level electronic health record (EHR) data to measure and reduce health disparities (Smith et al. 2023). Studies also assess consistency in hospital recommendations in Germany's report card system and the causes of ranking inconsistencies (Emmert et al. 2017). Evaluations of claims-based data's compliance with regulatory standards and its applications in hospital quality reporting and pay-for-performance programs have been conducted (Crews et al. 2016).

To **promote** accuracy and consistency in publicly reported infection data, standardized definitions and monitoring methods are recommended (Hebden 2012). Infection Composite Scores (ICS) have been developed to help hospitals assess infection control performance over time, identify underperforming areas, and implement improvements (Fakih et al. 2016).

Meanwhile, the use of data from the National Survey of Substance Abuse Treatment Services (N-SSATS) has been studied for its ability to capture additional information relevant to accessibility and service coordination (Vaughn et al. 2020).

System Usage

The role of the Public Health Automated Case Event Reporting (PACER) system and FHIR interoperability standards in improving public health reporting, especially during rapid infectious disease outbreaks, has been examined (Mishra et al. 2020). Studies have developed systems to track critical health outcome indicators in head and neck cancer treatment to enhance clinical practice and public reporting (Walters et al. 2014). When analyzing the rating distribution on the public reporting website Jameda, researchers examined how patient and physician characteristics influence overall scores (Emmert and Meier 2013).

Category	Key themes
Individual	• how public reporting influences individual healthcare choices • What cognitive factors, presentation formats, and patient characteristics influence individuals' understanding and use of public reporting
Organization	• how organizational attitudes shape responses to public reporting systems • This section examines how transparency and public reporting influence healthcare quality, efficiency, and patient-centered outcomes
Public reporting system	• how quality indicators influences systemic quality improvements • how performance indicators influence quality improvement across healthcare systems • efforts to improve the accuracy, consistency, and utility of data in public reporting • how interoperable system contribute to improved data exchange, outcome tracking, and rating accuracy

Table 9. *Classification of the main purposes of the public reporting systems studied in the reviewed articles.*

3.5 Consequences of the reviewed article in public reporting

3.5.1 Framework for Understanding Consequences

Results was classified to the formation of three themes: purposes related to individuals, purposes related to organization and purposes related to public reporting system.

For purposes related to individuals, we included evaluation tools and health literacy, while purposes related to organization involved organizational attitudes, regional disparities, and adverse impacts. Purposes related to public reporting system focused on quality indicators and standardized frameworks. Ultimately, we categorized the studies into seven themes discussed in the following section.

3.5.2 Individual-related consequences

Consumer Decision-Making

Studies indicate a correlation between hospital evaluations and traditional quality metrics. Highly rated hospitals on social media often exhibit better quality metrics, though this correlation is not absolute (Lagu and Greaves 2015). Low consistency in rankings among various report cards has also been observed (Emmert et al. 2017). Research suggests that physician rating websites significantly influence patients' physician selection, and this impact is likely to grow as public awareness increases (Emmert et al. 2013). A star rating system notably affects nursing homes' market shares, with consumers more likely to choose highly rated facilities (Werner et al. 2016). Similarly, public reporting influences consumer choices, with higher-rated home care facilities being more frequently chosen after the introduction of public reporting (Jung et al. 2016). Ratings for female doctors tend to be higher than for their male counterparts, and older patients and those with private health insurance generally provide higher ratings (Emmert and Meier 2013).

Health Literacy

Future public reporting developers should focus on user experience by creating reports tailored to different user groups (Shaller et al. 2014). Specific measures can increase minority awareness of public reporting (Yonek et al. 2014). Perceptions of public reporting are influenced by factors such as educational level, health status, and trust (Christianson et al. 2014). Patients with advanced dementia tend to perform poorly on specific quality indicators, particularly those related to functional decline and weight loss (Konetzka et al. 2015). MHCP patients generally score lower on quality indicators, with significant gaps persisting despite some improvements (Snowden AM et al. 2012). Results also indicate that users with no prior exposure to hospital report cards show a stronger intention to use them (Emmert and Wiener 2017).

3.5.3 Organization-related consequences

Stakeholder Perceptions

The availability of information, along with perceived value and attitudes of general practitioners, significantly impacts behavioral intentions, while perceived risk does not (Du, Zhang, et al. 2015). Acute care hospitals in Flanders have made substantial investments in quality improvement policies, with most hospitals completing initial certifications, publicly reporting quality indicators, and participating in pay-for-performance programs. Despite these efforts, stakeholders hold varying opinions on the effectiveness of these policies (Van Wilder et al. 2021).

Across Europe, public reporting of **HAI** data varies significantly. Experts in countries with established reporting systems generally support public reporting, while those in countries without such systems express reservations. Many experts advocate for reporting structure and process indicators over infection rates, citing concerns about data misinterpretation and biases (Martin M et al. 2013). Providers also worry that transparency could negatively affect their reputation and market share (Lihn et al. 2015).

Physicians often view online quality data on online resources with skepticism, doubting its accuracy. Only 2% consider Medicare's public reporting site highly accurate, and over half believe the data is not helpful for patients choosing doctors. They prefer foundational information, such as board certifications and clinical interests, over performance-based quality metrics or patient reviews (Lagu et al. 2019). Studies suggest that hospital report cards should emphasize physician and patient experiences to better support consultations. This implies that presenting hospital quality data requires adjustments to maximize its utility in clinical decision-making (Geraedts et al. 2018). The availability of information and the perceived value and attitudes of general practitioners significantly influence their behavioral intentions, whereas perceived risk does not exert a significant effect (Du, Zhang, et al. 2015).

Disparities Across Institutions

Evidence suggests that public reporting has had a significant impact on performance improvement (Poldrugovac et al. 2022). Public reporting transparency policies can enhance social welfare by reducing inequalities and encouraging hospital investments in quality improvement (Saghafian and Hopp 2020). The NPC-QIC initiative has achieved significant advancements in reducing infant mortality, improving nutrition management, and optimizing care processes (Lihn et al. 2015). In Wisconsin and California, public reporting systems have significantly improved healthcare service quality while reducing costs (Toussaint et al.

2014). However, while higher-performing hospitals have benefited from public reporting, lower-performing hospitals have not shown similar improvements (Bayindir and Schreyögg 2023).

Studies have also noted variation in improvements across institutions, with some metrics showing significant advancement while others have lagged (Jung K et al. 2010), performance data from public reporting (Lloyd et al. 2021). In one survey, 73% of PICU medical directors believed that benchmarking could improve patient outcomes (Natale et al. 2011). The impact of transparency is more pronounced for younger, wealthier, or urban patients compared to rural or older patients (Saghafian and Hopp 2020). Given the complex effects of public reporting on hospital efficiency across different market environments, policymakers are advised to consider market dynamics to achieve optimal healthcare outcomes (Han and Lee 2021). Studies also show that while some public reporting programs have improved clinical quality in specific healthcare domains, others primarily affect public perception rather than actual quality outcomes (Han and Park 2021). In specific cases, such as elective surgery for lung cancer, waiting times have slightly increased (Prang et al. 2021).

Un-intended consequences

Public reporting, while promoting transparency and accountability, can also result in unintended consequences, often referred to as “metric-driven harm” (Rambur et al. 2013). Patients generally support the public reporting of performance data but report increased anxiety, especially when they cannot change their healthcare provider (Duncanson et al. 2024). They prefer data to be presented positively, accompanied by explanations and educational resources to aid understanding and decision-making. Trust in their healthcare team remains paramount, with many patients relying more on physician advice than on statistical data (Duncanson et al. 2024).

In Germany, quality competition has improved healthcare standards, with quality measurement and public reporting recognized as key drivers of these improvements (Negele et al. 2023). Many studies recommend that regulators prioritize incentives for hospitals to improve quality, considering interventions that encourage hospitals to enhance their services (Saghafian and Hopp 2020). However, despite the transparency and accountability that public reporting promotes, unintended negative outcomes may arise. For instance, some physicians may avoid high-risk patients to improve their reported performance metrics. Furthermore, doctors and hospitals may focus more on meeting specific “target rates” rather than providing individualized care based on clinical judgment and patient needs. While some studies suggest that public reporting has contributed to lower cardiac surgery mortality rates, others find evidence that some physicians may refuse high-risk cases to maintain favorable evaluations. Additionally, while transparency has generally improved healthcare quality, studies reveal its impact can be limited, and in certain cases, it may even inadvertently reduce care quality (Werner and Asch 2005).

Financial burdens also accompany public reporting requirements (Wadhera et al. 2019). For example, hospitals performing percutaneous coronary interventions (PCI) face substantial costs, with catheterization labs incurring annual expenses ranging from \$100,000 to \$200,000, and in some cases, even higher (Lagu et al. 2019). Despite these challenges, combining incentive-based programs with patient experience reports and outcomes has shown promise in improving patient-centered care and delivering health outcomes that patients value (Schlesinger et al. 2015).

3.5.4 Public Reporting System -related consequences

Quality Indicator

Public reporting has been shown to positively impact quality indicators and performance indicators (Paddock 2014), such as reperfusion rates and AMI mortality (Chae et al. 2022), and reduce antibiotic prescription rates (Tang et al. 2017; Yang et al. 2014). Meanwhile, CLC quality improvements (Davila et al. 2024; Pimentel et al. 2021), vaccination rates (Paris et al. 2013), quality of care metrics (Hafner JM et al. 2011; Werner et al. 2009), infection control measures (Edmond and Bearman 2007; Fakhri et al. 2016), and cardiovascular care quality (Dehmer et al. 2014) have all improved following public reporting implementation. Substantial work is still needed for the content, design, and implementation of quality indicators (Hussey et al. 2014). However, public reporting may require adjustments to focus on factors patients find meaningful, such as hospital reputation and physician expertise, to increase their usage and impact (Ellis et al. 2020).

Data Processing Activities

Public reporting's influence on quality improvement has been limited, especially during the late 1990s, as third-party payers, patients, and referring physicians made little use of publicly available data for clinical decisions (Rhoads et al. 2009). A unified data processing framework could address these issues by providing an effective approach to measuring and monitoring health disparities (Smith et al. 2023). Collaborative efforts between public health agencies and healthcare systems have enabled faster case information collection, reduced human errors, and helped maintain data privacy and accuracy (Mishra et al. 2020).

Future systems should support automated, structured clinical data entry to effectively track and report treatment outcomes, thereby enhancing patient care and supporting a value-based healthcare system (Walters et al. 2014). Significant discrepancies in monitoring practices across institutions suggest the need for standardized definitions and automated systems to improve data accuracy and consistency (Hebden 2012). Standardized and validated reporting methods are also essential for improving report credibility and applicability (Crews et al. 2016; Johnson and Talbot 2011). Integration of patient experience data and real-time information is increasingly considered essential for the future of public reporting (Christianson et al. 2010). A new survey framework has been proposed to guide data collection and improve services in the future (Vaughn et al. 2020).

The format of data presentation in public reports, such as using charts or text and including confidence intervals, does not significantly impact consumer understanding (Mazor KM et al. 2009). However, providing detailed explanations within public reporting data enhances individuals' ability to comprehend and make informed choices (Duncanson et al. 2024). Supporting consumers in understanding performance data is essential. Research by Sander et al., (2015) identifies seven key elements that can improve user comprehension: bar charts without symbols, bar charts with simple symbols, bar charts with traffic light indicators, bar charts with thumb symbols, rankings of providers by performance, explicit statements indicating whether higher or lower values represent better performance, and performance ranges that suggest quality levels without explicitly stating scale direction. These elements—such as annotated bar charts, clear value indicators, and comparative rankings—translate complex data into accessible insights, thereby facilitating better public engagement with healthcare quality information.

3.6 A framework to summarize the research tradition of Public Reporting

This paper introduces the public reporting system as a third entity to examine its role in public reporting. The constructed framework diagram systematically elucidates the interactions among these three entities. Additionally, given the discrepancies between the purposes and results in the reviewed articles, this paper systematically explores and reports on this structure.

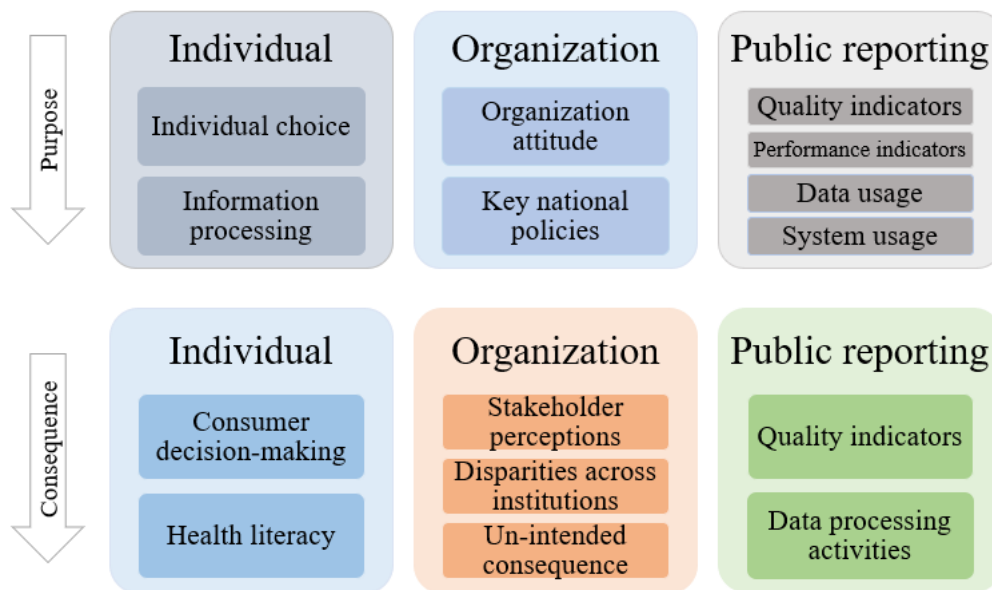


Figure 3. *A framework for understanding purpose and results in healthcare.*

The reviewed studies show that the purpose of supporting individual choice is reflected in actual consumer decision-making. Additionally, several articles conclude that health literacy significantly influences individual selection. It plays a key role in shaping healthcare access, patient-provider interactions, and self-care behaviors (Parnell 2014) .

The reviewed articles emphasize the importance of including patient experience reports and physician experience reports, reports should also place greater emphasis on factors that patient consider significant, such as hospital reputation and physician expertise (Ellis et al. 2020). Third, the exploration on the quality indicators has positive affect on the public reporting.

When it comes to the purpose is not aligned with the result, first, a number of evaluated research conclude that the purpose of organizational attitudes shows the stakeholder perception and organization disparities affect the public reporting implementation; yet, individual actions, based on rational choices, lead to organizational results (Coleman 1994).

While public reporting is essential for improving healthcare quality, it also presents challenges in two key areas. First, data-related issues can arise, as the way information is presented may cause anxiety among patients. Working towards data standardization and clarity is crucial, with careful consideration of individual factors and health literacy to make reports both accessible and informative.

Second, financial burdens pose another challenge. Compliance with public reporting requirements can place significant costs on healthcare institutions. Efficient resource allocation and policy adjustments are necessary to manage these financial pressures effectively.

4. Finding

Information published by public reporting systems consists of processed data, the processing directed at increasing its usefulness (Ackoff 1989; Totten et al. 2012). However, public reporting faces several challenges across data quality, performance measurement, and consumer engagement. From a data quality perspective, issues include the reliability of administrative claims data and time lags in public reporting, which diminish its relevance for real-time decision-making (Lihn et al. 2015). Potential biases may arise from manipulation or selective participation, further affecting data integrity (Emmert and Meier 2013).

In terms of performance data, challenges include inconsistencies in presentation formats, leading to difficulties in consumer interpretation (Hafner JM et al. 2011). Many consumers struggle to assess multiple provider performance measures without trusted guidance or simplified presentations (Shaller et al. 2014). Additionally, unintended consequences of public reporting, such as poorly adjusted data influencing provider behavior, may lead to risk-averse practices, such as avoiding high-risk patients to maintain favorable metrics (Rhoads et al. 2009).

Report cards introduce another layer of complexity, as providers may selectively report data or avoid high-risk patients to improve their rankings, resulting in biased samples (Werner and Asch 2005). Furthermore, discrepancies between reported data and actual quality can influence how report cards are perceived and utilized (Pimentel et al. 2021). Data biases have also been noted, particularly in pain assessments for dementia patients, where ascertainment issues affect reporting accuracy (Konetzka et al. 2015).

Electronic health records, despite their value in public reporting, pose challenges related to harmonization, incomplete data, and inconsistent collection methods (Smith et al. 2023). Public health data, while crucial, faces substantial barriers, including a lack of semantic interoperability that hinders data normalization across sources (Mishra et al. 2020). Variability in reporting methodologies across hospitals and countries further complicates benchmarking efforts (Martin M et al. 2013), (Johnson and Talbot 2011).

Patient-reported information presents additional obstacles, such as low patient engagement, fragmented data collection, insufficient coordination, and underrepresentation of diverse patient experiences (Lagu and Greaves 2015). The integration of data from various sources, such as patient narratives, complaints, and patient-reported outcome measures (PROMs), remains a significant challenge (Schlesinger et al. 2015).

Finally, challenges also arise from human factors, including data entry errors and misinterpretation of definitions, necessitating periodic manual corrections that add complexity to the reporting process (Paris et al. 2013). These findings underscore the multifaceted challenges in public reporting, highlighting the need for improved data standardization, risk adjustment, and enhanced usability to support the effectiveness of public reporting in healthcare.

This study systematically identified and reviewed 60 articles on public reporting in healthcare. Over time, public reporting has garnered increasing interest globally, with the Internet becoming the predominant way of developing public reporting, website information has been studied frequently, meanwhile, monitoring and prevention system and the new role of analysis, public reporting gained broad definition. Based on the findings, we will propose future research agendas and discuss research contributions and limitations in the following.

5. Discussion

Public reporting policies play a central role in shaping research contexts. A coding-based analysis of various studies revealed that countries implementing extensive public reporting policies, such as the United States, also produce the highest volume of related research. However, inconsistencies in national policies lead to significant variations in research backgrounds across different countries. Health technology assessment incorporates distinct traditions, practices, resources, and infrastructure, contributing to a multifaceted and evolving landscape (Fehr et al. 2017). To mitigate potential biases arising from these disparities, this study systematically categorizes and analyzes public reporting policies across countries, ensuring a comprehensive and balanced evaluation.

After 2015, countries like Austria, Belgium, and Korea emerged as countries with public reporting studied, indicating that public reporting has been developing globally. In the study by Emmert and Wiener (2017), a comparison was made between the United States and Germany. Although the research was primarily conducted in Germany, leading to the identification of these two countries in this article. Similarly, the study by Du, Zhang, et al. (2015) evaluated pay-for-performance programs in both the United States and Europe, adding another region to the overall count.

Traditionally, public reporting in healthcare was analog (paper-based). However, since 2015, there has been a significant shift towards digital (computerized). Globally, patient-centered public reporting systems incorporating user-generated data have become prevalent.

5.1 Future Agenda

5.1.1 Conceptual Agenda

Future research should prioritize developing a comprehensive and interdisciplinary conceptual understanding of public reporting. With the development of the digitalization, the definition of public reporting has expanded. Research in a single discipline led to the fragmentation within the findings in the literature, hindering the ability to compare and integrate findings effectively. Therefore, further research is required across multiple disciplines, particularly in fields such as health sciences, information systems, psychology, and computer science, to advance understanding and address the complexities of this topic.

5.1.2 Thematic Agenda

While previous research has predominantly concentrated on individual-related aspects and organization-related aspects, limited attention has been given to public reporting -related aspects in the healthcare context. Additionally, most studies have primarily examined the motivations behind specific individual side and organization side, with fewer studies addressing the barriers to effective public reporting practices. Moreover, while prior research has explored the outcomes of data usage and standard framework for providing an effective approach to measuring and monitoring health disparities, there remains a notable gap in understanding how data processes function across diverse healthcare contexts to achieve these outcomes. Based on these findings, we propose the following thematic agenda.

First, data serves as a powerful driver of economic growth worldwide (Statista 2022), among public reporting data, performance data and health records data rank as the most frequently reported categories. However, performance data often suffer from complexity and inconsistent presentation formats, making it difficult for consumers to interpret multiple provider performance metrics—especially in the absence of trusted guidance or simplified visualizations (Shaller et al. 2014). Additionally, publicly reported data, such as infection rates, may be misinterpreted or induce unnecessary patient anxiety, further complicating the effective use of public reporting for informed decision-making (Martin M et al. 2013).

Second, there is a need to explore information presentation in the public reporting. Research shows that information displays significantly impact user understanding, for example, bar graphs with symbols, clear markers of high and low values, and hospital rankings based on performance have been shown to enhance patient comprehension of public reporting data (Sander et al. 2015). In advancing personal systems, reports should also place greater emphasis on factors that patients consider significant (Ellis et al. 2020).

Third, there is a need to explore the stakeholder's attitude and actual behavior of public reporting. As digital media capabilities continue to advance, healthcare professionals increasingly require digital health literacy skills to integrate new information into their practices (Slevin et al. 2020). However, many physicians or experts maintain a skeptical or reserved stance on public reporting and use it less frequently (Van Wilder et al. 2021). **Given that organizations show improvement following the release of public reporting policy, future research should examine this from an organizational perspective, as organizations function as intermediaries between regulators and individuals.** Therefore, future research can focus on the stakeholder's attitude and actual behaviors.

Fourth, there is a need to explore the analysts as a user group. with RQ 1 come up with the monitoring system, analysts had shown up as a new user group in public reporting. Information accessibility and perceived value significantly influence healthcare professionals' behaviors (Du, Zhang, et al. 2015). Though European physicians primarily fulfill their information needs through online resources (Demergazzi et al. 2020), they may still rely on other sources for specialized or nuanced information. Monitoring system as new information resources need more study in the future research area.

5.1.3 Technological Agenda

Enhancing system interoperability is another critical focus for technological advancements in public reporting. Improved interoperability can increase data usability, boost operational efficiency, and enable rapid data sharing during infectious disease outbreaks. This helps address issues of data inconsistency across different platforms.

Data standardization and automation have been widely recognized in research as effective ways to improve data accuracy and consistency. Numerous studies emphasize the importance of using standardized definitions and automated systems to ensure reliable public reporting (Christianson et al. 2010; Crews et al. 2016; Hebden 2012; Johnson and Talbot 2011; Mishra et al. 2020; Smith et al. 2023; Vaughn et al. 2020; Walters et al. 2014).

Disparities in data storage systems also present significant challenges. Institutions use different structures—such as relational databases, NoSQL databases, or file systems—which creates fragmentation. Establishing a unified data standard across systems is essential, especially given variations in system updates, as well as differences between organizational and individual data storage practices.

Some studies also highlight significant variation in internal validation methods across healthcare institutions. These differences further underscore the need for consistent standards and validation protocols (Crews et al. 2016).

Finally, the implementation of public reporting systems varies considerably across European countries (Duncanson et al. 2024). This suggests that future research should focus on developing international strategies for data standardization and calibration to support cross-country comparability.

6. Contribution

6.1 Theory Implications

Health information systems play a crucial role in enhancing patient care (Walters et al. 2014). This study contributes to the field by systematically synthesizing, analyzing, and conceptualizing the current state of research on public reporting systems. First, through a structured explanation of public reporting system, it provides a framework by categorizing it into three key dimensions: individual, organizational, and public reporting structures.

Second, this study addresses a critical gap by bridging the disconnect between research objectives and unintended consequences in public reporting. Existing literature has identified that public reporting can lead to unintended consequences, resulting in users' distrust of the information sources (Li 2024; Moore et al. 2020; Theis et al. 2017). This study further explores how data transmission between systems contributes to data discrepancies and provides a system-level theoretical analysis of the unintended consequences of public reporting.

Third, while **pathway theorizing** has been extensively developed, the reputation pathway—one of its key components—has received limited attention since 2005. This pathway has the potential to enhance an organization's reputation (Hibbard et al. 2005), thereby increasing market share and fostering a mutually beneficial evolution within the healthcare organization.

Public perception of healthcare quality often diverges from actual clinical performance due to selective reporting and misinformation. By refining the mechanisms through which public reporting influences trust and credibility, healthcare institutions can leverage their reputation not only as a competitive advantage but also as a strategic tool for improving care quality and guiding patient decision-making.

6.2 Practical Implications

First, policymakers should consider market mechanisms when implementing public reporting to ensure optimal healthcare outcomes (Han and Lee 2021), with incentive structures that particularly address the needs of larger hospitals, as different public reporting programs have varying impacts on clinical quality (Han and Park 2021). Policy should integrate with market mechanisms to maximize the effectiveness of public reporting systems. The degree of improvement varies significantly (Hafner JM et al. 2011) may be due to increased emphasis on data by hospital leadership. Existing studies highlight the need for incentive policies, which could effectively enhance quality competition in the healthcare market. Policy making should focus on how these incentives can further encourage high-performing hospitals to improve healthcare outcomes. In this way, a patient-centered public reporting system can be established, enabling the timely collection of patient experiences and outcomes, promoting healthcare transparency.

Second, many practitioners lack formal training in medical coding or documentation improvement, and adherence to standards for physician queries and medical record corrections is suboptimal, with numerous institutions failing to follow prescribed timelines and correction procedures (Crews et al. 2016; Lagu et al. 2019; Martin M et al. 2013; Rhoads et al. 2009). By leveraging data for quality improvements, engaging healthcare staff in improvement activities, and clarifying roles and goals, these strategies indirectly contribute to achieving public reporting's quality improvement objectives (Davila et al. 2024).

Third, developers should consider the different stakeholders when developing the system, most public reporting system are directed toward organizational use, patients make up the largest group using public reporting systems, family members of patients also participate in these systems, system development should account for the needs of patient families, particularly in data-sharing options.

7. Conclusions and limitations

In this paper, a systematic review has been conducted of the literature based on the sample of 60 academic articles. The rise of patient-centered public reporting and user-generated data has expanded the scope of public reporting beyond patients, providers, and regulators to include analysts, monitoring systems, and digital health experts. This shift reflects the broader impact of digitalization in healthcare information dissemination and decision-making.

This study provides an up-to-date and comprehensive review of public reporting in healthcare, identifying key trends, challenges, and future research directions. The findings highlight the importance of standardization, usability, and digital transformation in public reporting. Addressing these challenges can enhance data transparency, stakeholder engagement, and healthcare quality outcomes. Future research should explore interdisciplinary approaches, stakeholder behaviors, and technological advancements to further improve public reporting systems in the healthcare sector.

This systematic literature review has certain limitations. First, as the study focuses specifically on public reporting in healthcare, the search strategy only included studies that explicitly mentioned public reporting. Consequently, relevant research exploring similar concepts without using this exact term may have been overlooked. Additionally, the review only included studies published in English, potentially omitting valuable insights from research in other languages.

Second, while this study categorizes public reporting within individual, organizational, and system-level frameworks, we acknowledge that alternative classifications may also provide valuable insights.

This systematic literature review examines public reporting, analyzing its evolution, impact, and future potential. By exploring the relationships between public reporting systems, data components, and key stakeholders, this study provides valuable insights into the effectiveness of public reporting in driving healthcare transparency and quality enhancement.

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Appendix A. Concept matrix of the purpose of the reviewed articles.

Lloyd et al.(2021)			√			√		√				√			
Han & Park(2021)				√								√			
Han & Lee(2021)				√								√			
Vaughn et al.(2020)							√	√							√
Saghafian & Hopp(2020)				√								√	√		
Mishra et al.(2020)								√							√
Ellis et al.(2020)					√									√	
Wadhera et al.(2019)			√					√					√		
Lagu et al.(2019)			√								√		√		
Geraedts et al.(2018)						√					√				
Tang et al.(2017)						√								√	
Emmert & Wiener(2017)	√							√		√					

Emmert et al.(2017)							√		√						
Werner et al.(2016)	√								√						
Jung et al.(2016)	√								√						
Fakih et al.(2016)							√							√	
Crews et al.(2016)							√								√
Zhang et al.(2015)			√								√				
Schlesinger et al.(2015)				√									√		
Sander et al.(2015)		√													√
Lihn et al.(2015)				√							√	√			
Lagu & Greaves(2015)	√								√						
Konetzka et al.(2015)		√								√					
Du et al.(2015)				√							√	√			

Yonek et al.(2014)		√								√					
Yang et al.(2014)					√									√	
Walters et al.(2014)								√							√
Toussaint et al.(2014)					√							√			
Shaller et al.(2014)		√								√					
Hussey et al.(2014)	√													√	
Paddock (2014)						√								√	
Dehmer et al.(2014)					√									√	
Christianson et al.(2014)		√								√					
Rambur et al.(2013)			√										√		
Paris et al.(2013)					√									√	
Martin et al.(2013)			√								√				

[illegible]